Unit 4 - Important Topics

Feed systems in liquid propellant rockets

21ASC305T - Rocket Propulsion
Slides ke basis par simple Hinglish me samjhaya hua notes

Unit 4 - Important Topics (Liquid Rocket)

Ye PDF sirf liquid propellant rockets ke feed systems ka hai. Har topic ke end me **Exam Answer** (English me, full marks ke liye) diya hai.

Is PDF me kya kya hai

- Feed systems in liquid propellant rockets: functions, gas pressure feed vs turbopump feed
- Gas pressure feed systems: regulated aur blow-down (comparison table ke saath)

Source ka matlab: hare box me PDF ka naam aur page number (slide number) diya hai. Feed systems wala poora part slides se hai (UNIT 4 RP - L2.pdf, Page 4 se 15). Unit 4 ke slides me koi numerical nahi hai.

1. Propellant feed systems

Source: UNIT 4 RP - L2.pdf | Page 4 se 8 (functions p4, selection p5, gas vs turbopump p6, diagram p7, turbopump tank pressure p8)

Feed system ke 2 principal functions hain:

- Propellants ka pressure badhana.
- Unhe design mass flow rate par ek ya zyada thrust chambers tak supply karna.

In kaamo ki energy ya to **high-pressure gas** se aati hai, ya **centrifugal pumps** se, ya dono ke combination se.

Feed system ka selection kis par depend karta hai

- Rocket application aur mission
- Duration
- Thrust chambers ki number ya type
- Past experience
- Design ki simplicity, manufacturing ki aasani, low cost aur minimum inert mass

Gas pressure feed vs turbopump feed

Gas pressure feed systems turbopump systems se **behtar** performance dete hain jab:

- Vehicle ka total impulse ya propellant mass relatively low ho.
- Chamber pressure relatively low ho.
- Engine thrust-to-weight ratio low ho (usually 0.6 se kam).

Gas pressure system me heavy-walled propellant tanks aur pressurizing gas hi engine system ka major inert mass hote hain.

Turbopump feed system me propellant tank pressure **10 se 40 guna kam** hota hai, aur isliye vehicle ke tank masses bhi usi factor se kam ho jaate hain.

Diagram (Page 7): regulated gas pressure feed system ki schematic. Labels: gas tank, gas start valve, pressure regulator, fuel tank aur oxidizer tank, fuel main valve, oxidizer main valve, injector, combustion chamber, exhaust nozzle, check valves, vent and relief valves, isolation valves, fill and drain valves.

	Gas pressure feed	Turbopump feed
Kab behtar	Low total impulse, low chamber pressure, low thrust-to-weight (< 0.6)	Bade impulse aur high chamber pressure ke liye
Tank pressure	High (tank ko heavy-walled banana padta hai)	10 se 40 guna kam
Tank mass	Zyada	Kaafi kam (usi factor se)
Energy source	High-pressure gas	Centrifugal pumps (turbine driven)

EXAM ANSWER (write this in English for full marks)

Q. Explain the propellant feed systems of a liquid propellant rocket engine.

A propellant feed system has two principal functions: (1) to raise the pressure of the propellants and (2) to supply them at the design mass flow rate to one or more thrust chambers. The energy comes from a high-pressure gas, from centrifugal pumps, or from a combination of the two.

The choice of feed system depends on the rocket application, duration, number and type of thrust chambers, past experience, mission, and the general requirements of simple design, easy manufacture, low cost and minimum inert mass.

Gas pressure feed system versus turbopump feed system

- A gas pressure feed system gives better performance when the total impulse or propellant mass is low, the chamber pressure is low and the engine thrust-to-weight ratio is low (usually less than 0.6).
- In a gas pressure feed system the heavy-walled propellant tanks and the pressurizing gas form the major inert mass of the engine.
- In a turbopump feed system the propellant tank pressures are lower by a factor of 10 to 40, so the tank masses are lower by the same factor.

(Draw the labelled schematic of a gas pressure feed system: gas tank, gas start valve, pressure regulator, fuel and oxidizer tanks, propellant valves, injector and thrust chamber.)

2. Gas pressure feed systems

Source: UNIT 4 RP - L2.pdf | Page 9 se 15 (types p9, regulated components p10, diagram p11, scavenging p12, blow-down p13-14, comparison table p15)

Liquid propellants ko tanks se bahar dhakelne ka sabse simple aur common tareeka high-pressure gas hai. Aaj bhi do common types use hote hain:

Type 1: Regulated gas pressure feed system

- Gas feed line me ek **gas pressure regulator** hota hai. Isse engine lagbhag constant tank pressure par aur lagbhag constant thrust par chalta hai.
- Main components: high-pressure gas tank, gas starting valve, gas pressure regulator, propellant tanks, propellant valves aur feed lines.

Extra components bhi ho sakte hain: filling aur draining provisions, check valves, filters, flexible elastic bladders (liquid ko pressurizing gas se alag rakhne ke liye), pressure sensors ya gauges.

- Jab propellants poore consume ho jaate hain to pressurizing gas ka use lines aur valves se bacha hua liquid propellant residue saaf karne (scavenge) ke liye bhi kar sakte hain.
- Agar unit ko baar baar use ya fly karna ho (jaise space-maneuver rocket) to usme kuch extra features ho sakte hain, jaise thrust-regulating device aur tank level gauge.

Type 2: Blow-down feed system

- Propellant tanks **bade** hote hain kyunki unme propellants ke saath pressurizing gas bhi initial maximum tank pressure par store hoti hai.
- Alag high-pressure gas tank aur pressure regulator **nahi** hote.
- Tank ke andar pehle se bhari gas ka expansion hi propellants ko bahar dhakelta hai. (Isliye tank pressure aur thrust dheere dheere girte jaate hain.)

Diagram (Page 11): regulated system ka detailed layout: high-pressure gas supply tank, gas bleed valve, pressure regulator, tank vent valves, filters, check valves, oxidizer tank aur fuel tank, propellant valves (remote control), restricting orifice, rocket thrust chamber aur optional additional thrust chamber(s).

	Regulated system	Blow-down system
Gas storage	Alag high-pressure gas tank	Propellant tank ke andar hi gas
Regulator	Hota hai	Nahi hota
Tank size	Propellant tanks chhote	Propellant tanks bade
Tank pressure / thrust	Lagbhag constant	Burn ke saath girta hai
Complexity	Zyada parts	Simple

Slide ki comparison table (Page 15)

	Regulated pressure	Blowdown
Pressure / thrust	Essentially constant	Propellant consume hone ke saath decrease hota hai
Gas storage	Alag high-pressure tanks me	Gas propellant tank ke andar hi stored (tank pressure par, large ullage volume 30-60%)
Required components	Regulator, filter, gas valves aur gas tank chahiye	Bade, bhaari propellant tanks
Advantages	Nearly constant-pressure feed se essentially constant propellant flow aur approximately constant thrust, Isp aur mixture ratio. Mixture ratio ka behtar control.	Simpler system. Kam gas chahiye. Inert mass kam ho sakta hai. Koi high-pressure gas tank nahi.
Disadvantages	Thoda zyada complex. Regulator se chhota pressure drop. Gas high pressure par lambe time tak stored rehti hai. Zyada pressurizing gas chahiye.	Thrust burning duration ke saath girta hai. Mixture ratio ka control kam accurate hone se thoda zyada residual propellant. Thruster ko thrust values ki wide range aur mixture ratio ki modest range me stable chalna padta hai. Propellants pressure me stored; burning time ke end par Isp thoda kam.

EXAM ANSWER (write this in English for full marks)

Q. Describe the gas pressure feed systems (regulated and blow-down types) used in liquid rockets.

Using a high-pressure gas is one of the simplest and most common ways to push liquid propellants out of their tanks. Two types are in common use.

1. Regulated gas pressure feed system

- A gas pressure regulator is placed in the gas feed line, so the engine works at essentially constant tank pressure and nearly constant thrust.
- **Main components:** high-pressure gas tank, gas starting valve, gas pressure regulator, propellant tanks, propellant valves and feed lines. Additional parts may be filling and draining provisions, check valves, filters, flexible elastic bladders to separate the liquid from the gas, and pressure sensors or gauges.
- After the propellant is used up, the pressurizing gas can scavenge and clean the lines and valves of residue. Reusable units may also have a thrust-regulating device and a tank level gauge.

2. Blow-down feed system

- The propellant tanks are larger because they store the propellants and also the pressurizing gas, at the initial maximum tank pressure.
- There is no separate high-pressure gas tank and no pressure regulator. The expansion of the gas already in the tanks pushes out the propellants.
- Monopropellant gas pressure feed systems are simpler because there is only one propellant, so fewer pipes, valves and tanks are needed.

Comparison

- **Pressure/thrust:** regulated - essentially constant; blow-down - decreases as the propellant is consumed.
- **Gas storage:** regulated - separate high-pressure tanks; blow-down - inside the propellant tanks (large ullage volume, 30 to 60%).
- **Components:** regulated - regulator, filter, gas valves and gas tank; blow-down - larger, heavier propellant tanks.
- **Advantages:** regulated - nearly constant propellant flow, thrust, I_{sp} and mixture ratio, with better mixture-ratio control; blow-down - simpler system, less gas required, can have less inert mass, no high-pressure gas tank.
- **Disadvantages:** regulated - slightly more complex, the regulator causes a small pressure drop, the gas is stored at high pressure often for a long time, and more pressurizing gas is needed; blow-down - thrust decreases with burn duration, slightly higher residual propellant due to less accurate mixture-ratio control, the thruster must work over a wide range of thrust and a modest range of mixture ratio, and I_{sp} is slightly lower towards the end of burning.